

**9.2.33.**

Six people attend the theater together and sit in a row with exactly six seats.

- a. How many ways can they be seated together in the row?
- b. Suppose one of the six is a doctor who must sit on the aisle in case she is paged. How many ways can the people be seated together in the row with the doctor in an aisle seat?
- c. Suppose the six people consist of three married couples and each couple wants to sit together with the older partner on the left. How many ways can the six be seated together in the row?

- a. How many ways can they be seated together in the row?

There're six people and six available seats. Hence:

$$P(6, 6) = \frac{6!}{(6-6)!} = \frac{6!}{0!} = \frac{6!}{1} = 6!$$

**Solution: a.** There're  $6!$  ways the six people can be seated together in the row.

- b. Suppose one of the six is a doctor who must sit on the aisle in case she is paged. How many ways can the people be seated together in the row with the doctor in an aisle seat?

The doctor has two choices, either the seat furthest to the left or the seat further to the right. Once the doctor has chosen one of those two seats, there are five people and five seats remaining. Hence:

$$P(5, 5) = \frac{5!}{(5-5)!} = \frac{5!}{0!} = \frac{5!}{1} = 5!$$

But we must also account for the doctor's two possible choices of seats. Therefore:

$$2 \times P(5, 5) = 2 \times 5!$$

**Solution: b.** There're  $2 \times 5!$  ways the people can be seated in the row with the doctor in an aisle seat.

c. Suppose the six people consist of three married couples and each couple wants to sit together with the older partner on the left. How many ways can the six be seated together in the row?

We have three married couples, and each couple must sit together. So we are effectively arranging three objects, and since the order within each couple is already fixed, it's a 3-permutation with three elements. Hence:

$$P(3, 3) = \frac{3!}{(3-3)!} = \frac{3!}{0!} = \frac{3!}{1} = 3! = 6$$

**Solution:** c. There're 6 ways the people can be seated in the row, if each couple wants to sit together.